

$$\sigma = -\text{real}(\lambda) = 0.554492 \text{ sec}^{-1}$$

$$99\% \text{ damping time} = \frac{\ln(0.01)}{-\sigma} = \frac{\ln(0.01)}{-0.554492 \text{ sec}^{-1}} = 8.305 \text{ sec}$$

$$\omega_d = |\text{imag}(\lambda)| = 3.090658 \text{ rad/sec}$$

$$\text{period} = \frac{2\pi}{\omega_d} = \frac{2\pi}{3.090658 \text{ sec}^{-1}} = 2.033 \text{ sec}$$

Following Example 8.1.1, the amplitudes for the four components of this pair of eigenvectors are

$$\text{amplitude} \begin{Bmatrix} \Delta V_1 \\ \Delta V_2 \\ \Delta x_1 \\ \Delta x_2 \end{Bmatrix} = C_{1,2} \begin{Bmatrix} 0.659219 \text{ ft/sec} \\ 0.688002 \text{ ft/sec} \\ 0.209942 \text{ ft} \\ 0.219109 \text{ ft} \end{Bmatrix}$$

and the phase shifts relative to ΔV_2 are

$$\text{phase} \begin{Bmatrix} \Delta V_1 \\ \Delta V_2 \\ \Delta x_1 \\ \Delta x_2 \end{Bmatrix} = \pm \begin{Bmatrix} -2.9045 \\ 0.0000 \\ 1.6251 \\ -1.7464 \end{Bmatrix} = \pm \begin{Bmatrix} -166.42^\circ \\ 0.00^\circ \\ 93.11^\circ \\ -100.06^\circ \end{Bmatrix}$$

Notice that this mode has changed very little from mode 1 in Example 8.1.1. This is because the mode primarily involves the vibration of only the second spring, which has not changed.

The undamped natural frequency for this mode is

$$\begin{aligned} \omega_n &= \sqrt{\lambda_1 \lambda_2} = \sqrt{(-0.554492 + 3.090658 i)(-0.554492 - 3.090658 i)} \\ &= \sqrt{0.554492^2 + 3.090658^2} = 3.140004 \text{ rad/sec} \end{aligned}$$

and the damping ratio is

$$\begin{aligned} \zeta &= -\frac{\lambda_1 + \lambda_2}{2\sqrt{\lambda_1 \lambda_2}} = -\frac{-0.554492 + 3.090658 i - 0.554492 - 3.090658 i}{2\sqrt{(-0.554492 + 3.090658 i)(-0.554492 - 3.090658 i)}} \\ &= \frac{0.554492}{\sqrt{0.554492^2 + 3.090658^2}} = 0.176590 \end{aligned}$$